Deep Generative Models

Lecture 2

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- Outline
  1. Autoregressive models
  2. Normalizing Flow models
  3. Diffusion models
     - Advanced Image editing
  4. Summary and comparison

Taxonomy of Deep Generative Models

- Several approaches:
  1. Variational Autoencoders [Kingma-2014]
  2. Generative Adversarial Networks (GANs) [Goodfellow-2014]
  3. **Autoregressive models** [Oord-2016]
  4. **Normalizing flow models** [Dinh-2017]
  5. **Diffusion models** [Sohl-Dickstein-2015, Rombach-2022]

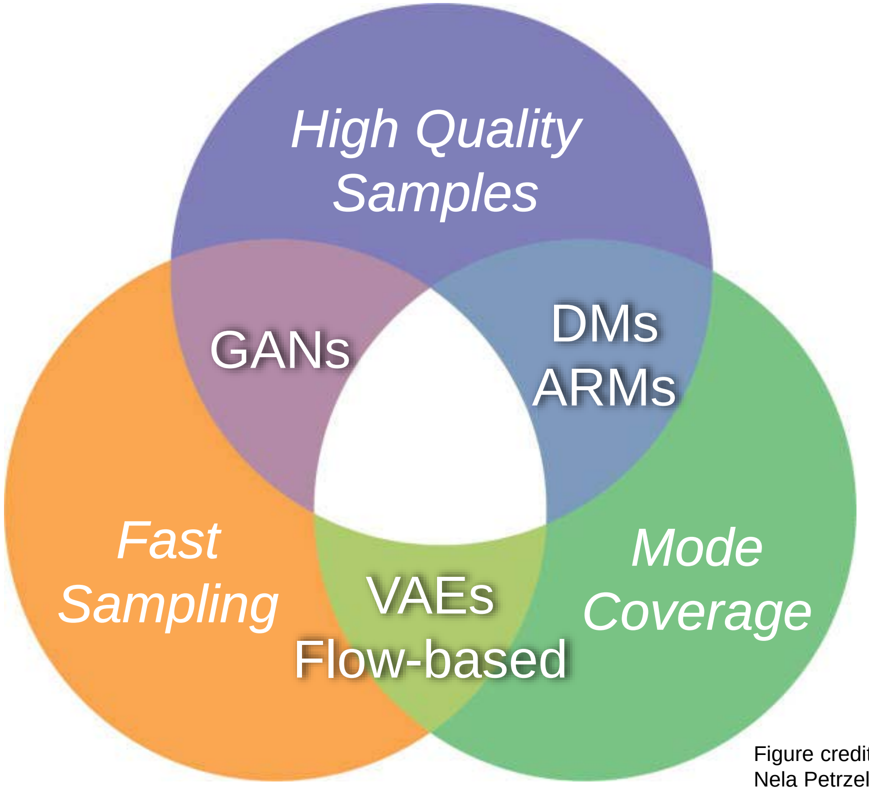


Figure credit
Nela Petrzelkova

Autoregressive Models

Autoregressive models

- Data likelihood modelled as

$$p(\mathbf{x}) = \prod_{t=1}^{T} p(x_t \mid x_1, \ldots, x_{t-1})$$

- Sampling: iterative sequential generation

$$\begin{aligned} x_1 &\sim p(x_1) \\ x_2 &\sim p(x_2 \mid x_1) \\ x_3 &\sim p(x_3 \mid x_1, x_2) \\ &\vdots \\ x_T &\sim p(x_T \mid x_1, \ldots, x_{T-1}) \end{aligned}$$

"Token by token, pixel by pixel"

- Training – maximum likelihood

$$\mathcal{L} = \sum_{t=1}^{T} \log p_\theta(x_t \mid x_1, \ldots, x_{t-1})$$

  – No adversarial loss, no variational bound
  – Exact likelihood (in theory)

Autoregressive models

PROS:

- Simple to train
  - Straightforward maximum likelihood
- High-quality samples
- Exact data likelihood

CONS:

- Sampling is slow
- No latent space, No inversion available

- Use cases:
  - **State-of-the-art for text generation** (GPT models)

Normalizing Flow Models

Normalizing Flow models

- Normalizing flow
  - **Invertible** neural transformation that maps data x into latent space z

$$z = f_\theta(x) \qquad z \sim p_Z(z) = \mathcal{N}(0, \mathbf{I})$$ "latent distribution"

$$x = f_\theta^{-1}(z) \qquad x \sim p_X(x)$$ "distribution of data"

- Easy sampling
- Exact likelihood (change of variable formula)

$$p_X(x) = p_Z(f_\theta(x)) \left| \det \frac{df_\theta(x)}{dx} \right|$$

- Training via maximum (log-)likelihood

$$\max_\theta \sum_{i=1}^{N} \log p_Z(f_\theta(x_i)) + \log \left| \det \frac{df_\theta(x_i)}{dx_i} \right|$$

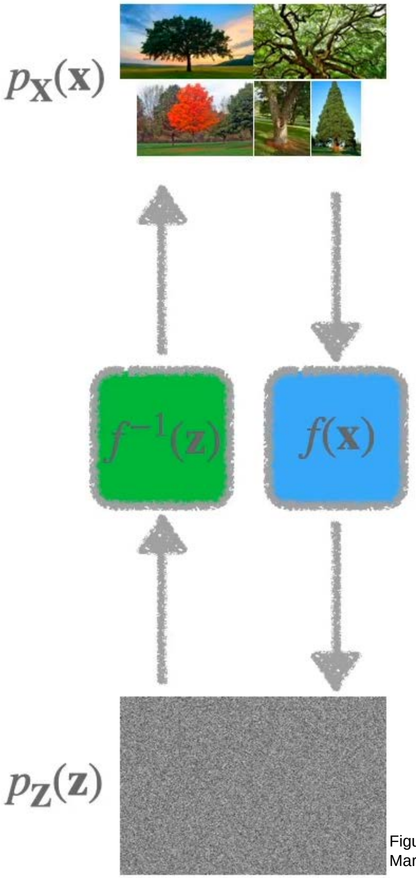


Figure credit: Marcus Brubaker

Normalizing Flow models

- But, how to find the flow function $f_\theta(x)$?
  - Construct the complex mapping out of simple functions ~ neural net

$$f = f_K \circ f_{K-1} \circ \ldots \circ f_1$$

  - Each layer must be:
    - Invertible
    - Differentiable
    - Cheap Jacobian to compute

"composition of simpler flows"

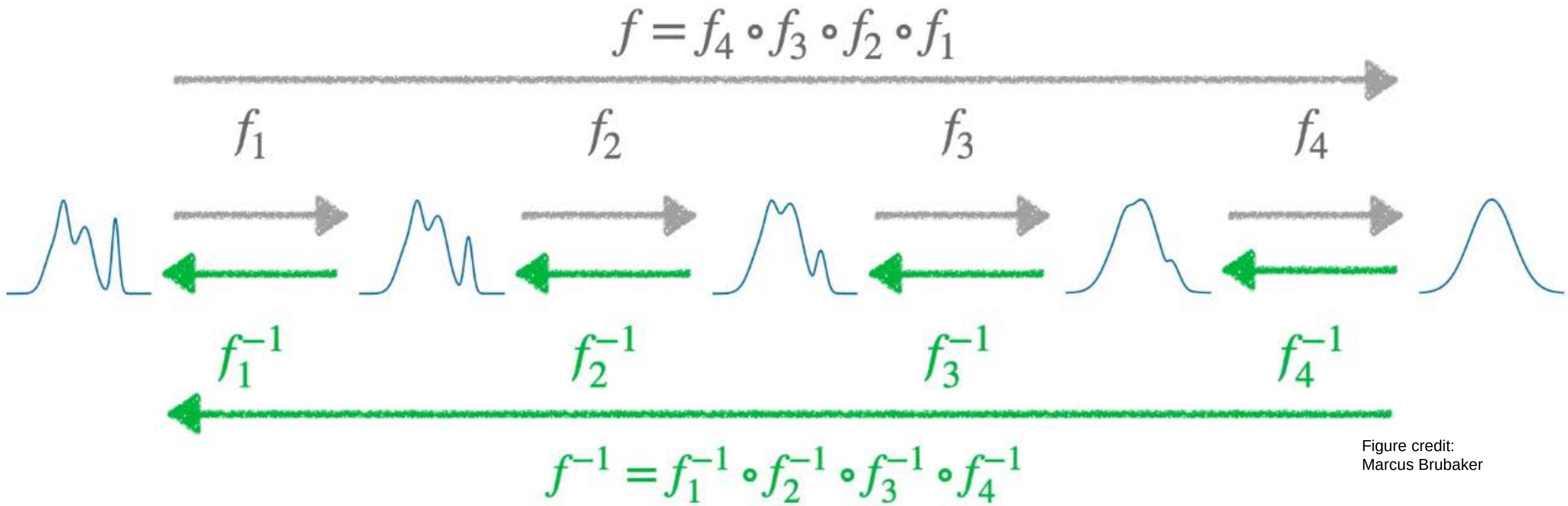


Figure credit:
Marcus Brubaker

Normalizing Flow models

PROS:

- **Exact inversion**
- Simple and fast sampling
- Exact data likelihood
- Straightforward maximum likelihood training
- High-quality samples (recently)
- Meaningful latent space (interpolation, semantic directions)

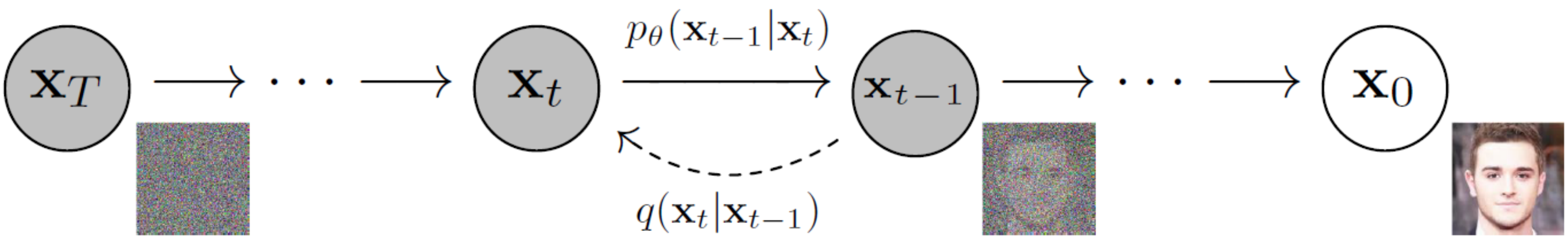
CONS:

- Dimensionality of latent space (same as the pixel space)
- Memory heavy
- Expressivity limited by invertibility constraint

Diffusion Models

Diffusion Models

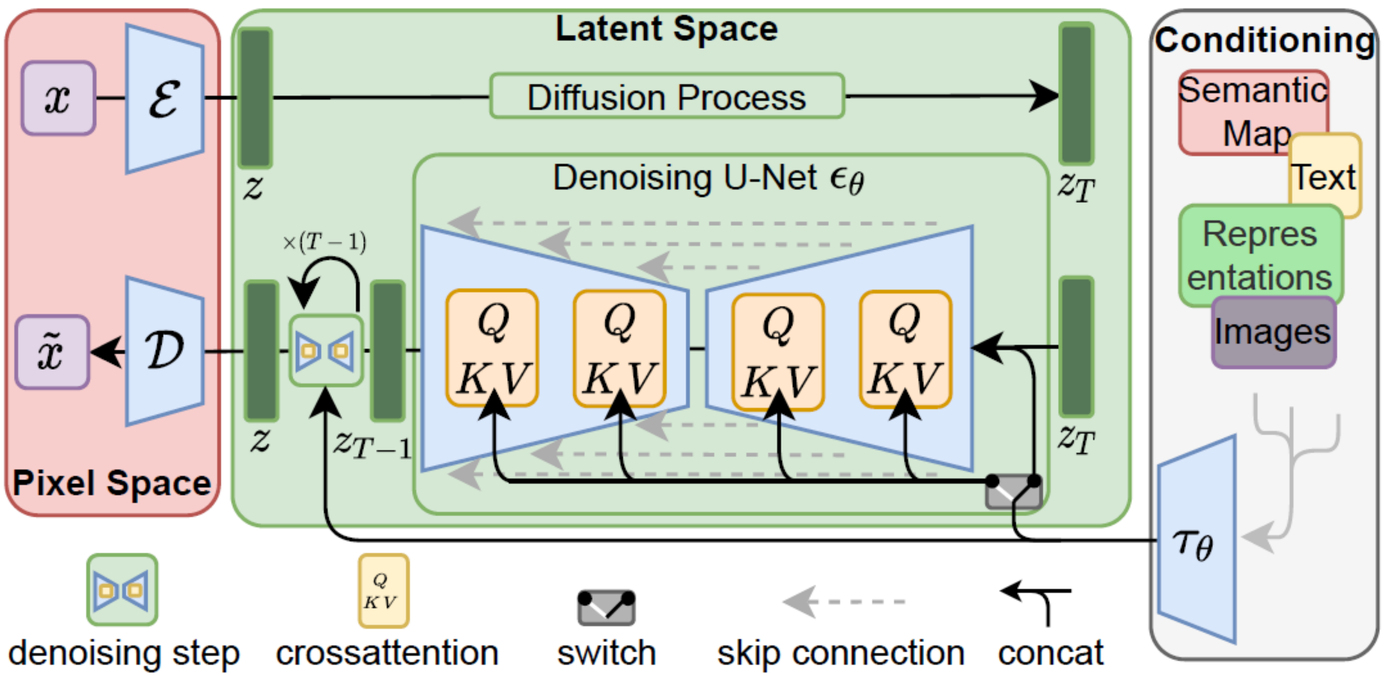
- Inspiration from thermodynamics [Sohl-Dickstein-2015]
- Denoising diffusion probabilistic models [Ho-2020]
- Main principle:
  - Forward diffusion: progressively destroy the data by injecting noise
    - Gaussian Noise
  - Reverse diffusion: learn to reverse the process to sample generation
    - denoising U-NET



$$L_{DM} = \mathbb{E}_{x,\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right]$$

Stable Diffusion

- Stable Diffusion [Rombach-2022]
  - Latent diffusion model (diffusion/denoising runs in latent space)
  - Encoder/decoder pixel-latent space learnt offline
  - Conditioning by cross-attention (Text, Semantic maps, …)

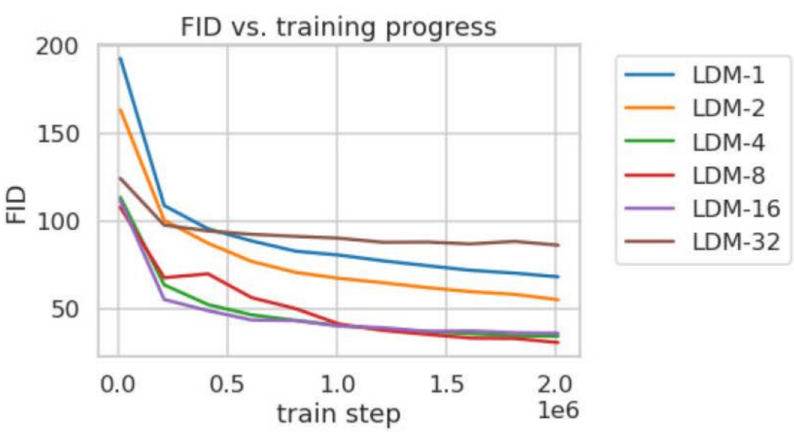


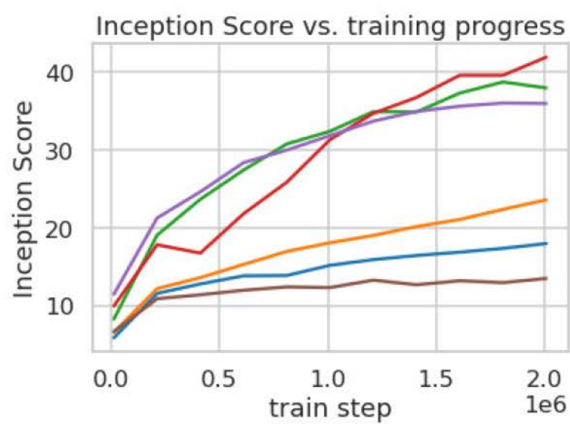
$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x),y,\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon-\epsilon_\theta(z_t,t,\tau_\theta(y))\|_2^2\right]$$

Stable Diffusion

- Unconditioned generation (256x256)

  – Effect of the latent “compression”





LDM-1
(pixel level)

LDM-8
(default)

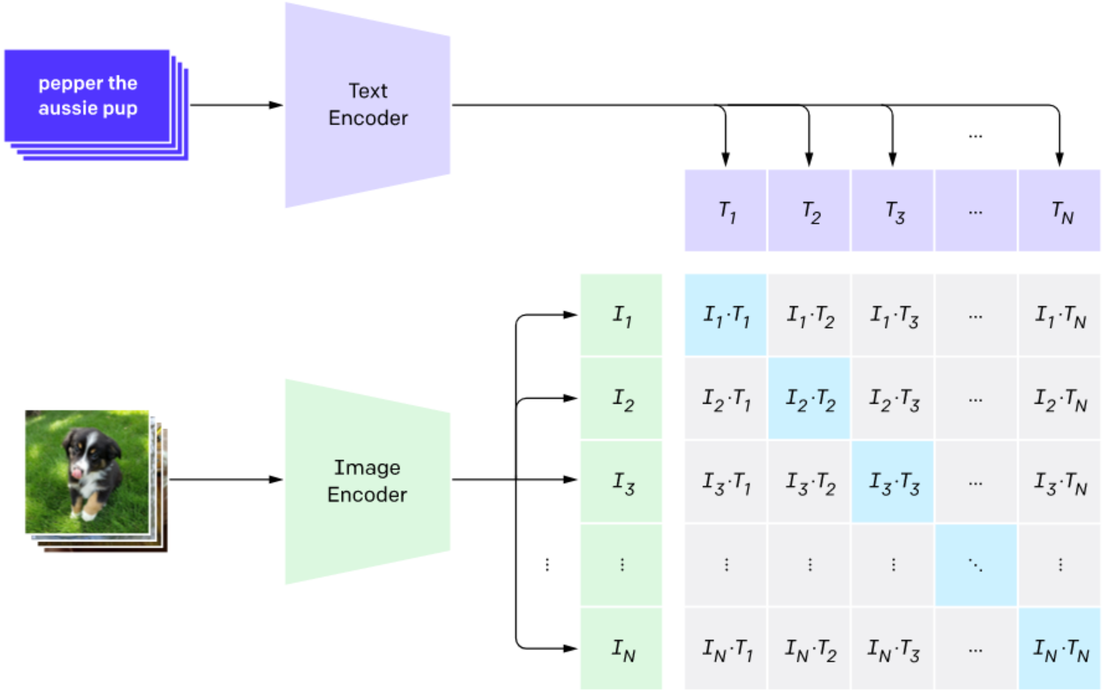
Stable Diffusion

- Text-to-Image
  - Trained on LAION dataset - 1.45B
  - CLIP embedding for text

'A street sign that reads "Latent Diffusion" ' | *'A zombie in the style of Picasso'* | *'An image of an animal half mouse half octopus'* | *'An illustration of a slightly conscious neural network'* | *'A painting of a squirrel eating a burger'* | *'A watercolor painting of a chair that looks like an octopus'* | *'A shirt with the inscription: "I love generative models!" '*

CLIP – Connecting Text and Images (recap)

- CLIP [Radford-2021] by OpenAI
  - *"Contrastive Language–Image Pre-training"*
  - Learn joint text-image embedding => Text-image (cosine) similarity
  - Learned from 400M WebImageText (WIT) dataset



  - Zero-shot prediction (on par with Resnet on ImageNET benchmark)
    - Loop over ImageNET-classes: max CLIP(E_T(*"A photo of a <class>"*), $E_I(I)$)
  - Trained model publicly available

Stable Diffusion

- Semantic maps

Semantic Synthesis on Flickr-Landscapes [23]

- Inpainting

input | result

Classifier-Free Guidance (CFG)

- Classifier-free diffusion guidance [Ho-2022]
- Controls adherence with the input prompt
  - Trade-off mode coverage and sample fidelity

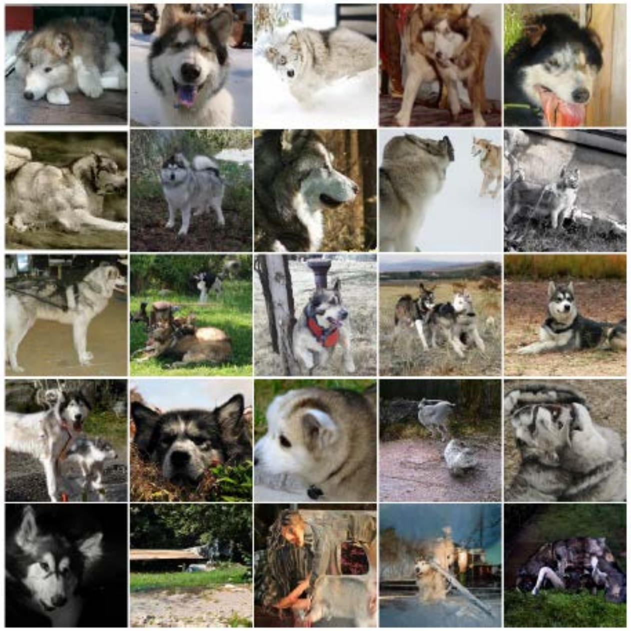
- Training: Diffusion model is trained both conditional and unconditional
  - Unconditional sample (null prompt) is given in certain probability p_{uncond} (e.g. 0.1/0.2/0.5)

- Inference:
  - Sample a weighted combination of conditioned and unconditional denoising model in each step

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_\lambda, \mathbf{c}) = (1 + w)\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - w\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda)$$

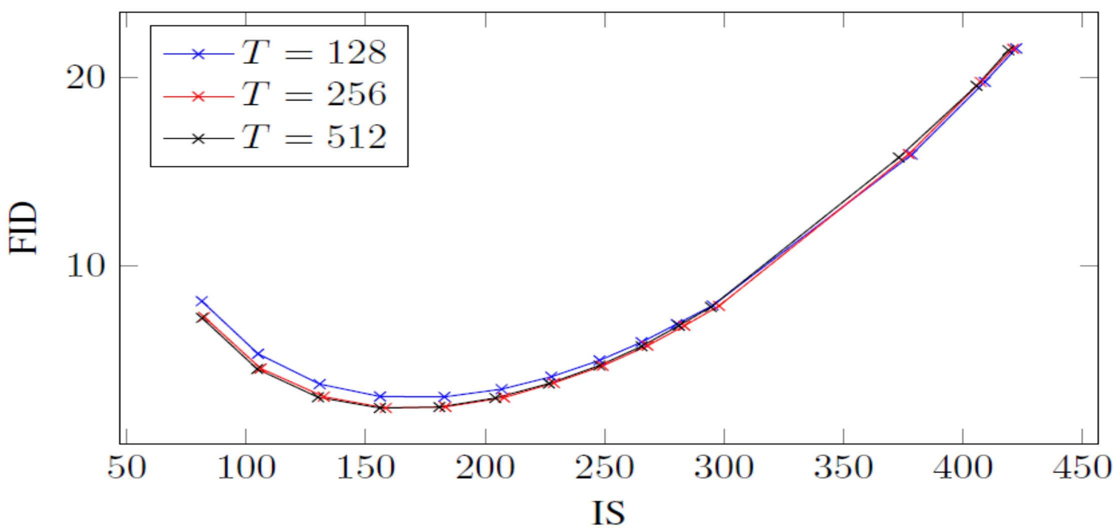
Classifier-Free Guidance (CFG)

$w = 0$



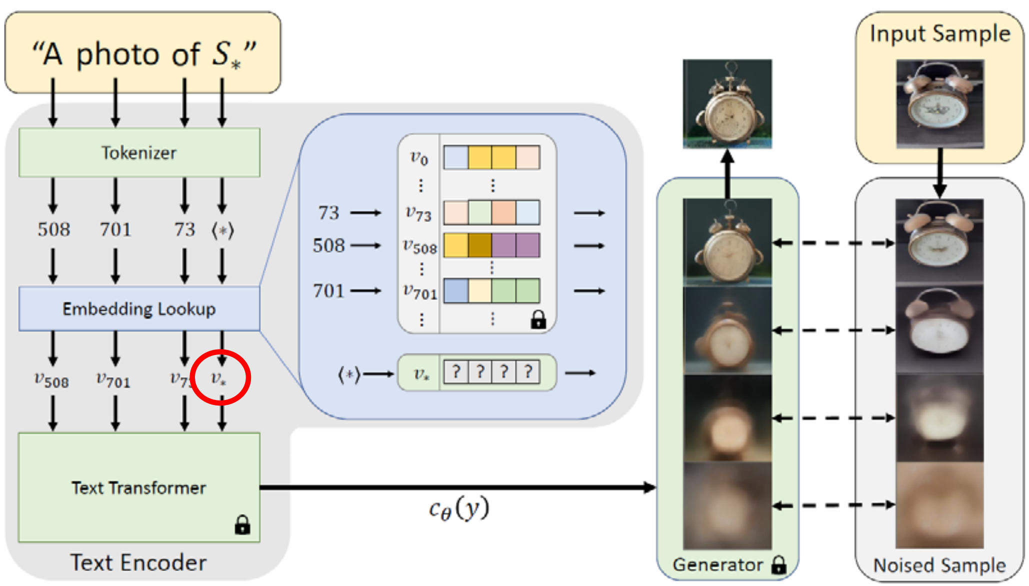
$w = 1$

$w = 4$



Text2image using Textual Inversion

- An image is worth one word... [Gal-2022]
- Text to image with custom objects

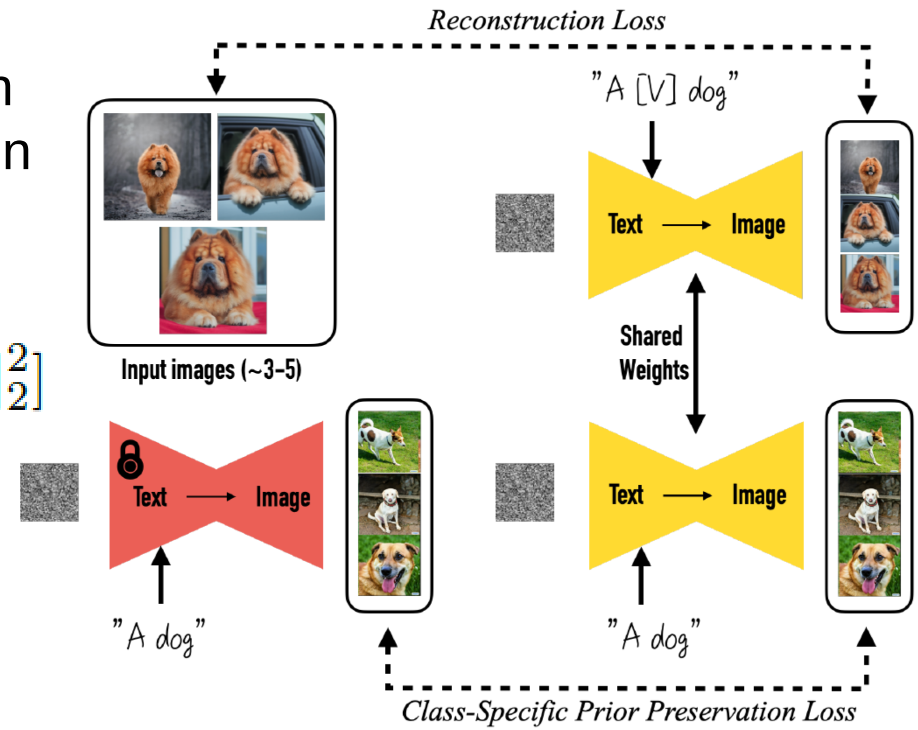


$$v_* = \arg\min_v \mathbb{E}_{z\sim\mathcal{E}(x),y,\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(z_t, t, c_\theta(y))\|_2^2\right]$$

DreamBooth

- DreamBooth [Ruiz-2022] (Google)
- **Fine-Tuning** Text-to-Image Diffusion Models for Subject-Driven Generation

$$\mathbb{E}_{\mathbf{x},\mathbf{c},\epsilon,\epsilon',t}\left[w_t \|\hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x} + \sigma_t \epsilon, \mathbf{c}) - \mathbf{x}\|_2^2 + \lambda w_{t'} \|\hat{\mathbf{x}}_\theta(\alpha_{t'} \mathbf{x}_{\text{pr}} + \sigma_{t'} \epsilon', \mathbf{c}_{\text{pr}}) - \mathbf{x}_{\text{pr}}\|_2^2\right]$$



  – about 5 mins on A100 GPU

Fine-tuning Diffusion Models

- LoRA (Low-Rank Adaptation) [Hu-2021] (Microsoft)
  - General light-weight adaptation of any models (including LLMs)
    - Faster training and storing
  - Many models available (e.g., CivitAI, PromptHero)

$$\mathbf{W}' = \mathbf{W} + \mathbf{AB}$$

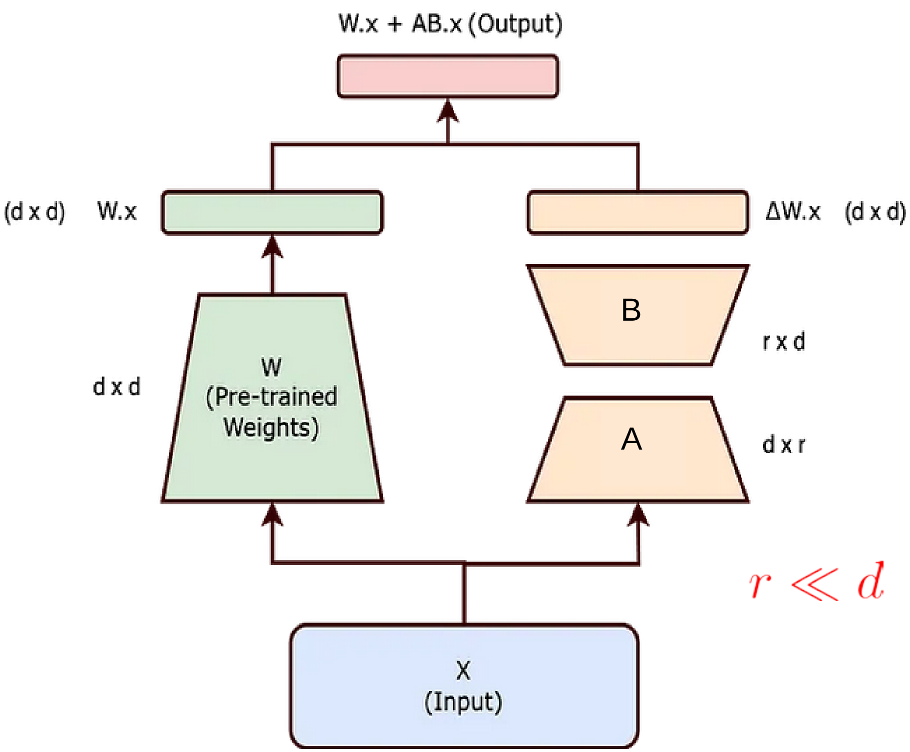
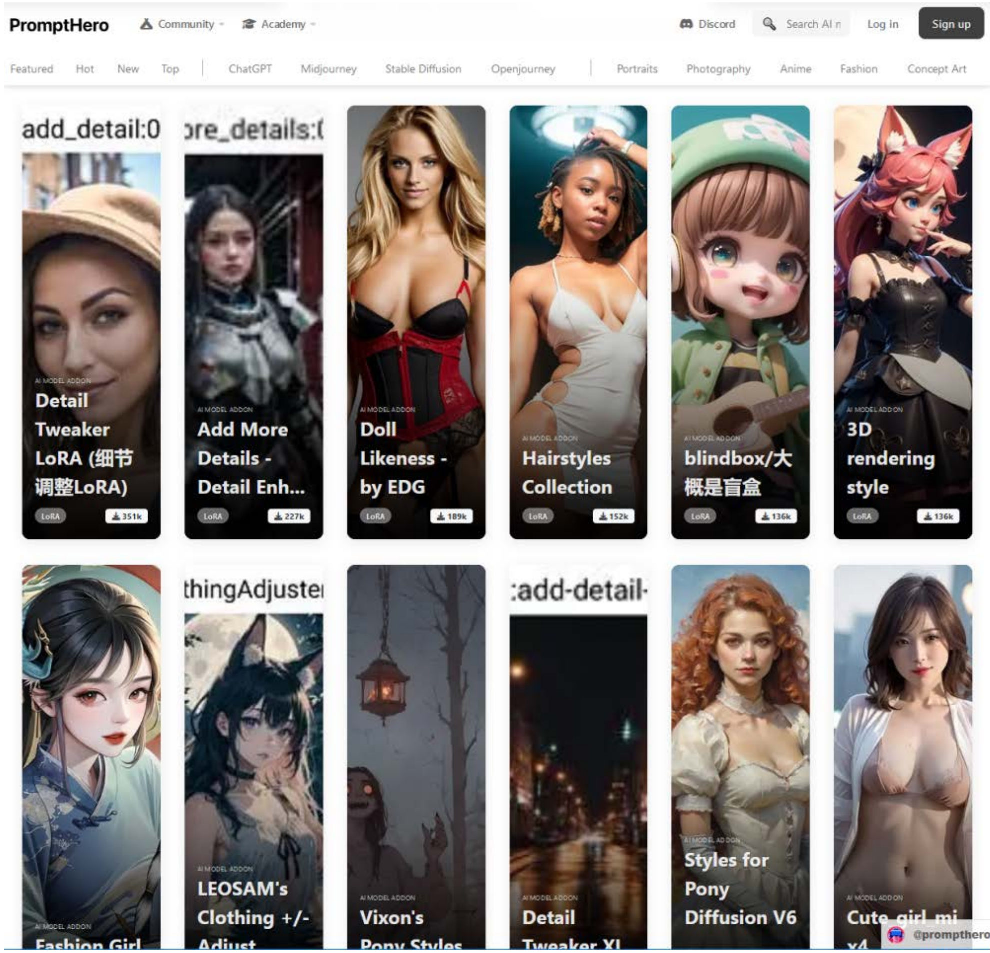
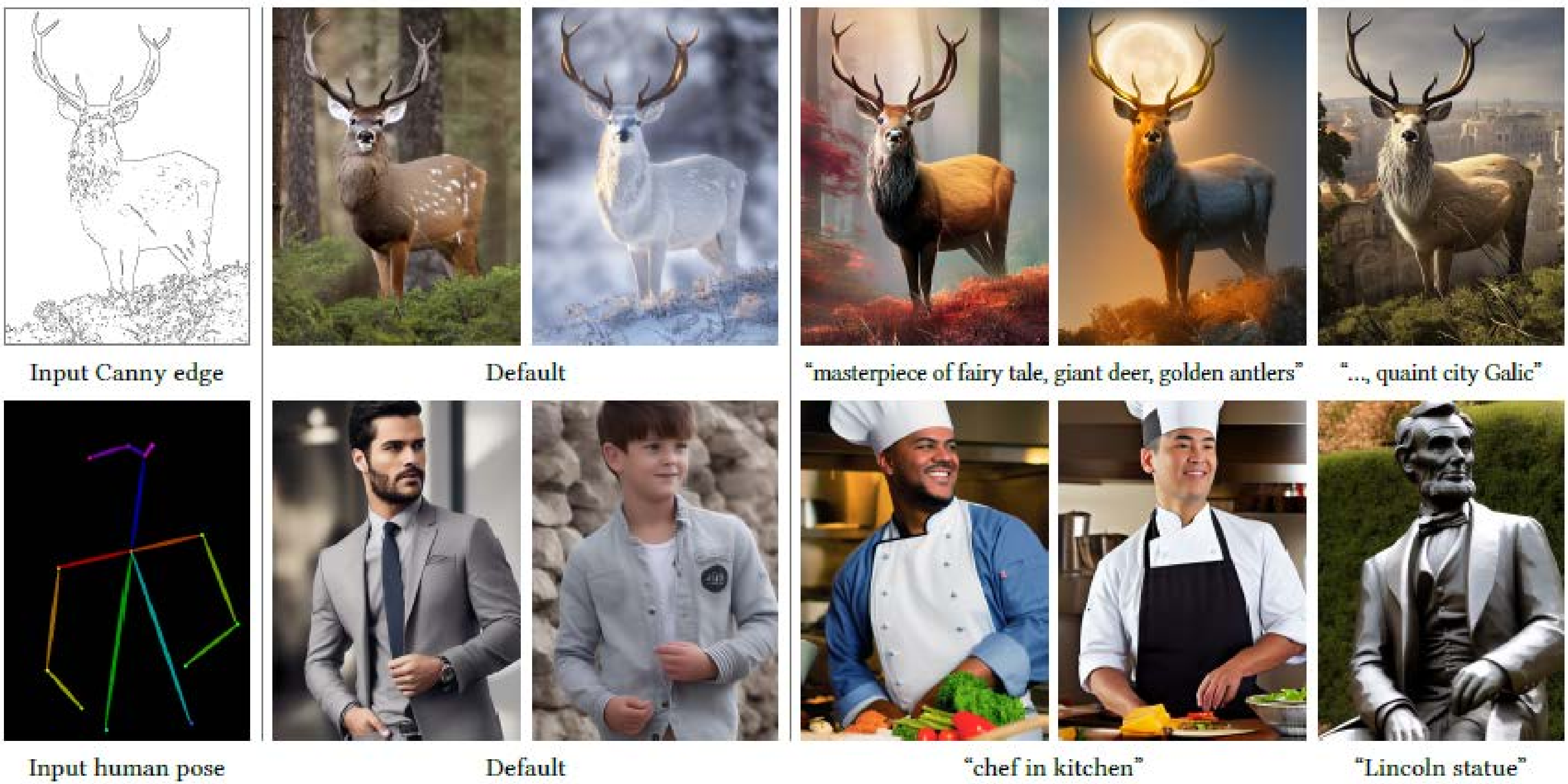


Figure credit Bhavin Jawade



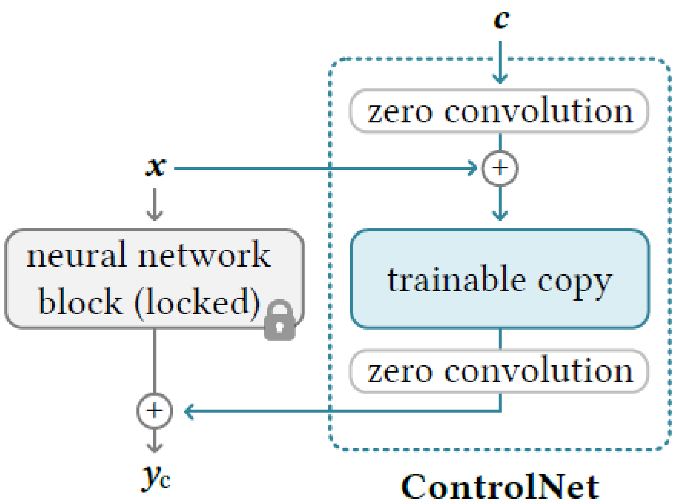
ControlNet

- Adding Conditional Control to Text-to-Image Diffusion Models [Zhang-2023]



ControlNet

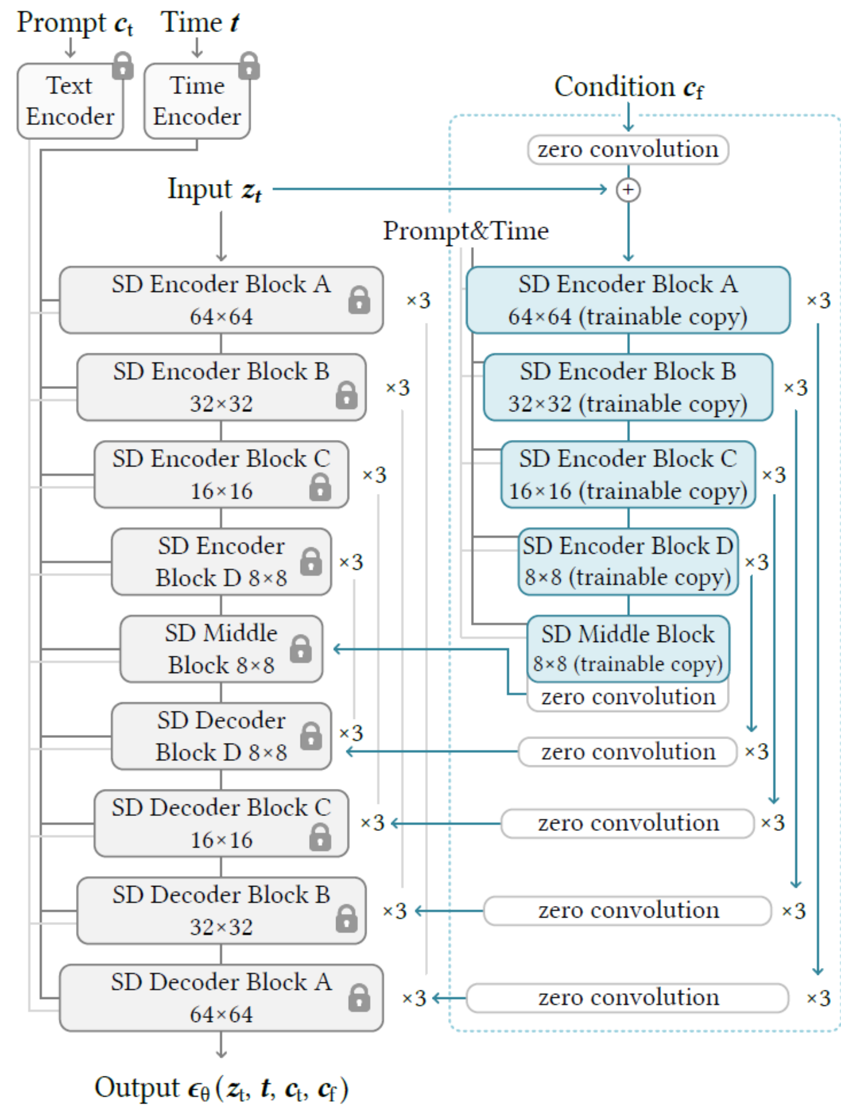
- Fine-tuning with a trainable copy
  - Zero convolutions: 1x1 convolutions with weights initialized to zeros



Extra cond.

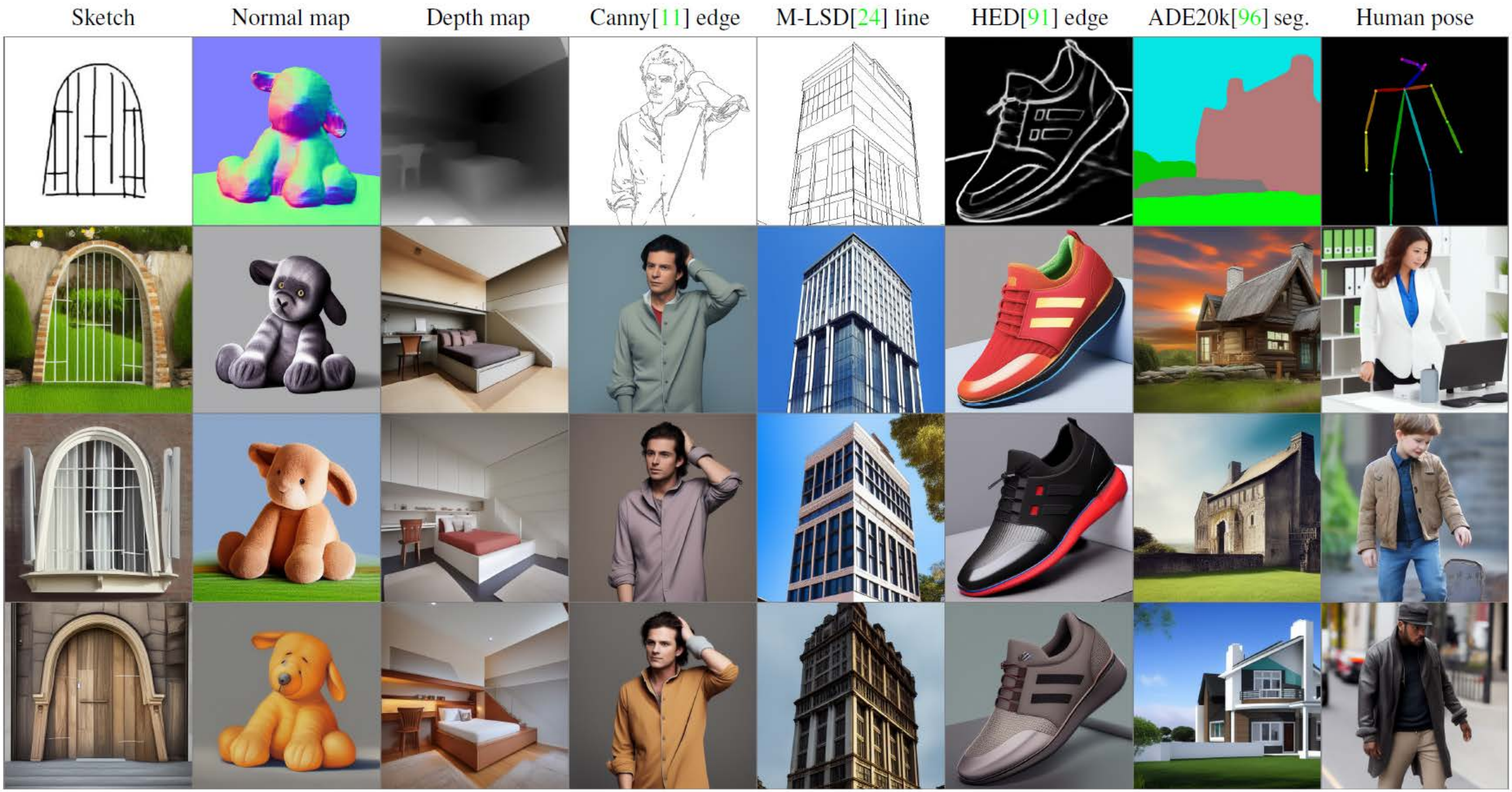
$$\mathcal{L} = \mathbb{E}_{\boldsymbol{z}_0, \boldsymbol{t}, \boldsymbol{c}_t, \boldsymbol{c}_f, \epsilon\sim\mathcal{N}(0,1)}\left[\|\epsilon - \epsilon_\theta(\boldsymbol{z}_t, \boldsymbol{t}, \boldsymbol{c}_t, \boldsymbol{c}_f)\|_2^2\right]$$

  - 50% of text prompt c_t randomly replaced with empty string



ControlNet

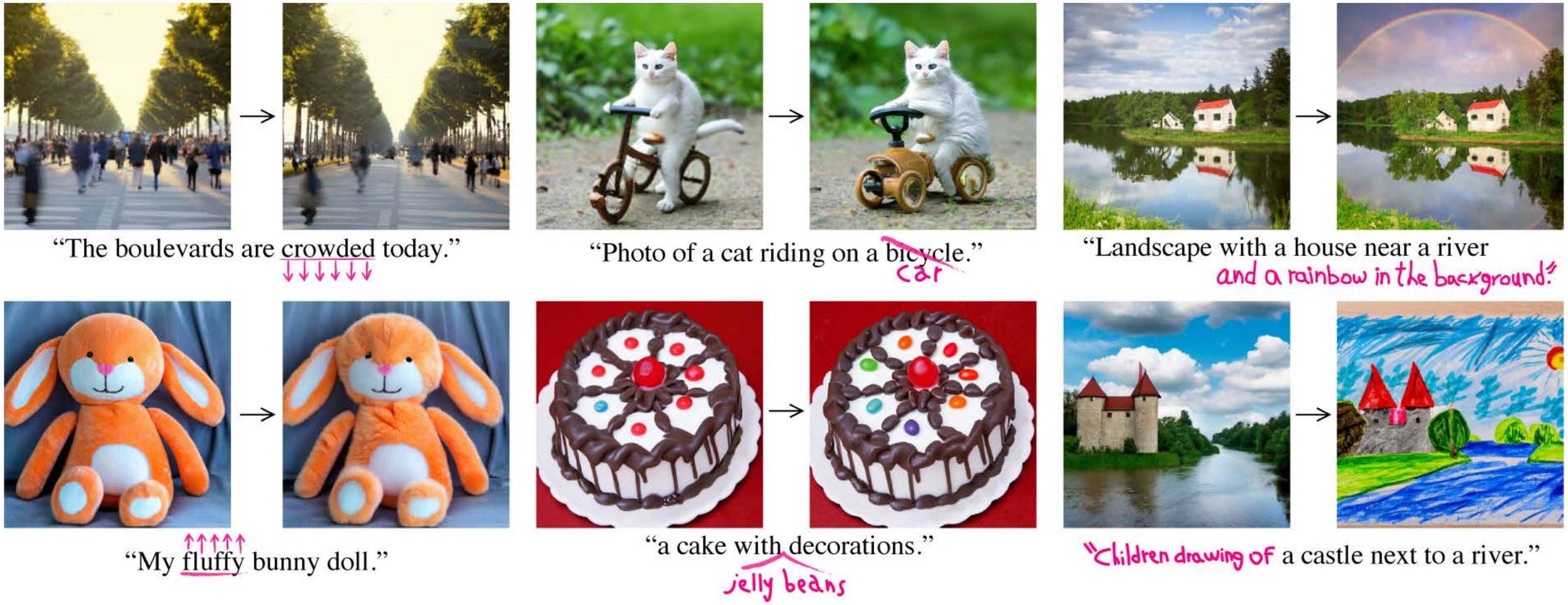
- Models trained for multiple conditioning



- Demo, code and models available

Prompt2Prompt

- Prompt-to-Prompt image editing [Hertz-2022]



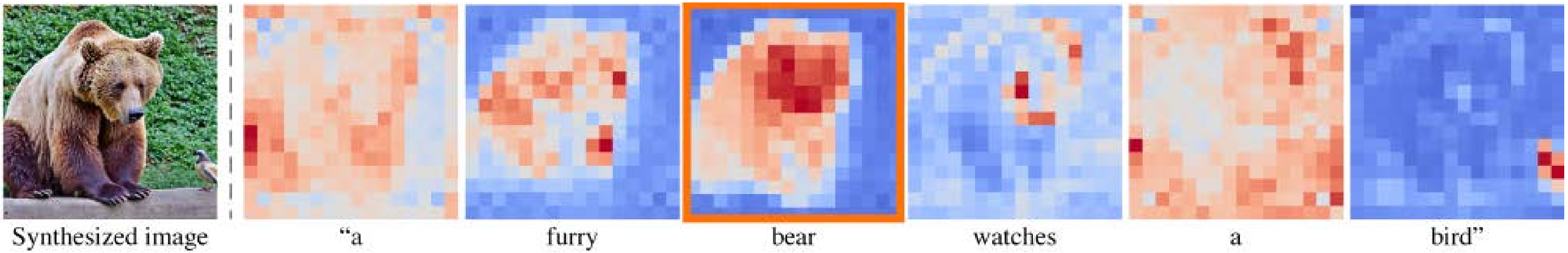
fixed random seed

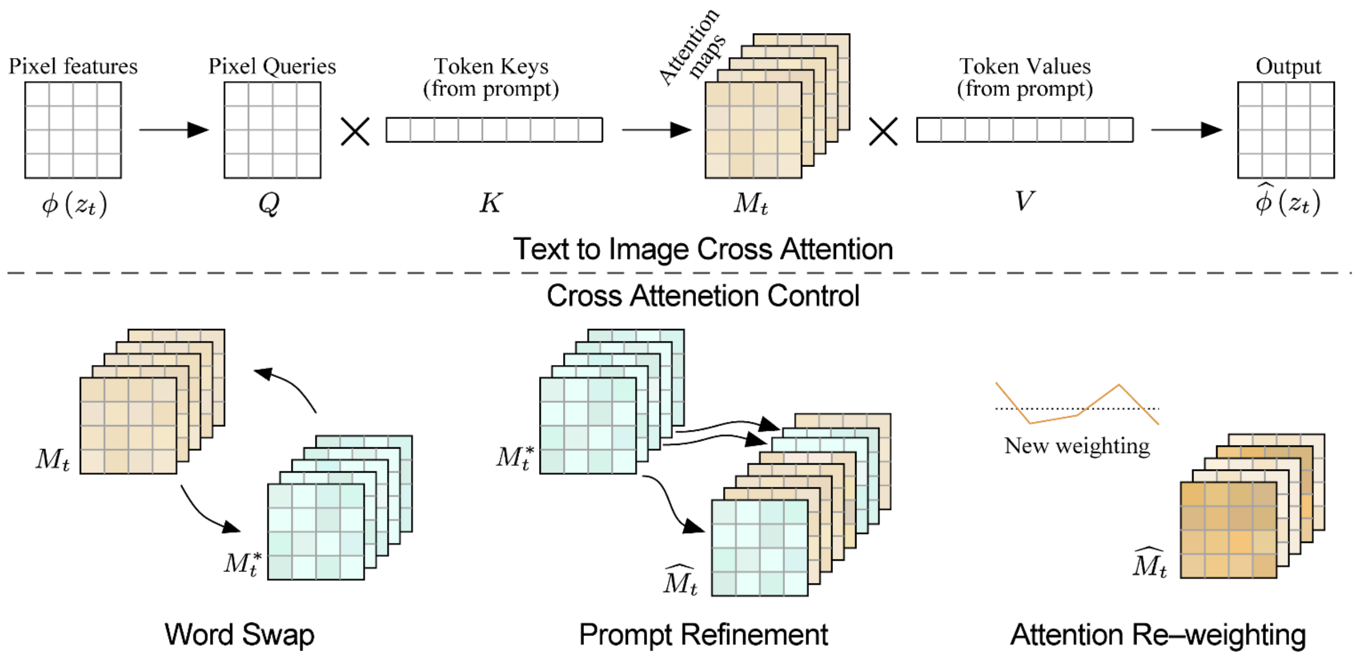
fixed random seed and attention maps

  - Generated image editing
    - Stress/Weaken words, Changing words, Adding new phrases
    - Fixing random seed does not help (layout changes)

Prompt2Prompt

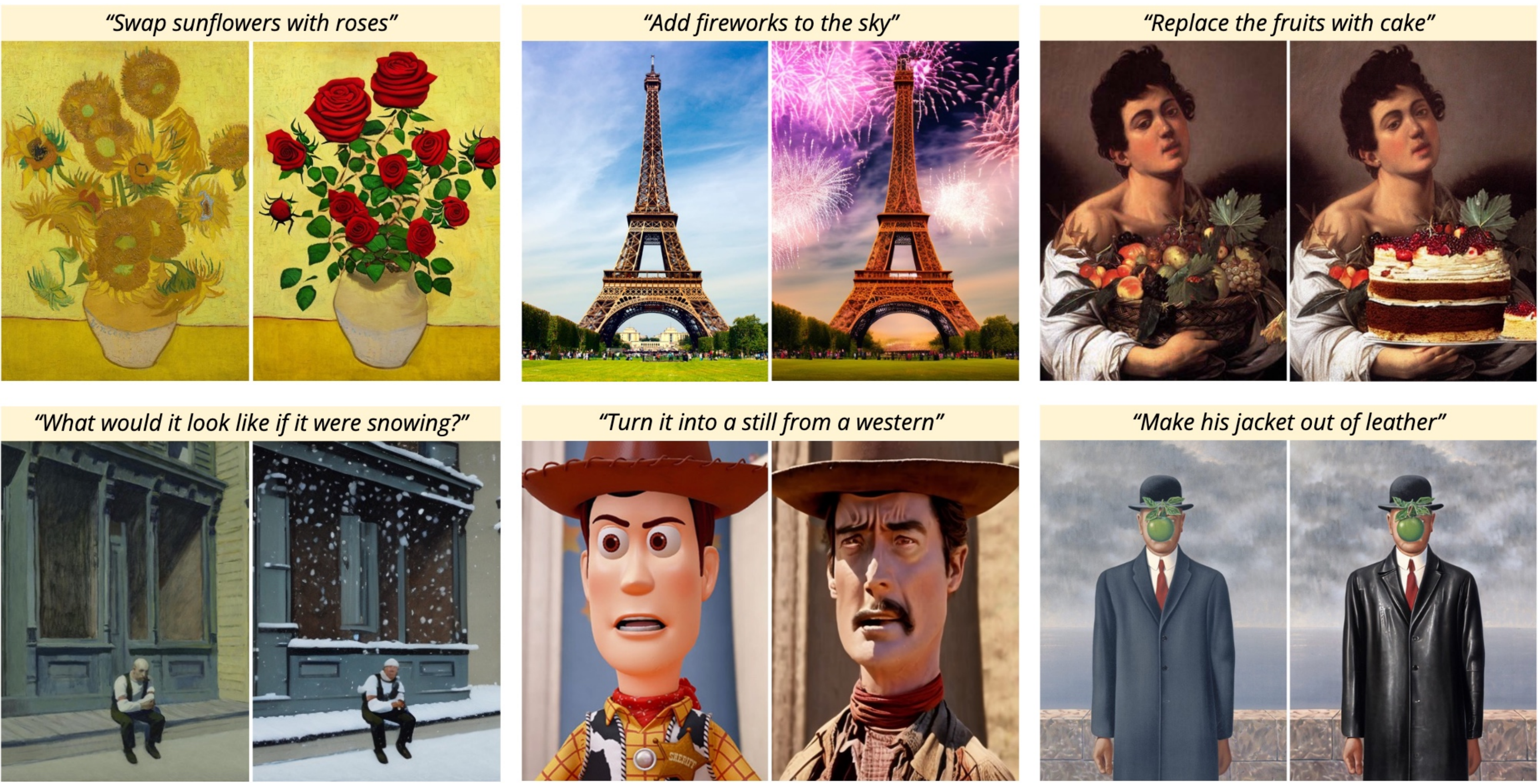
- Controlling the Cross-Attention (text x image)
  - Attention maps are responsible for spatial layout
  - Keep existing attention maps, change/add new word maps, reweight





Instruct Pix2Pix

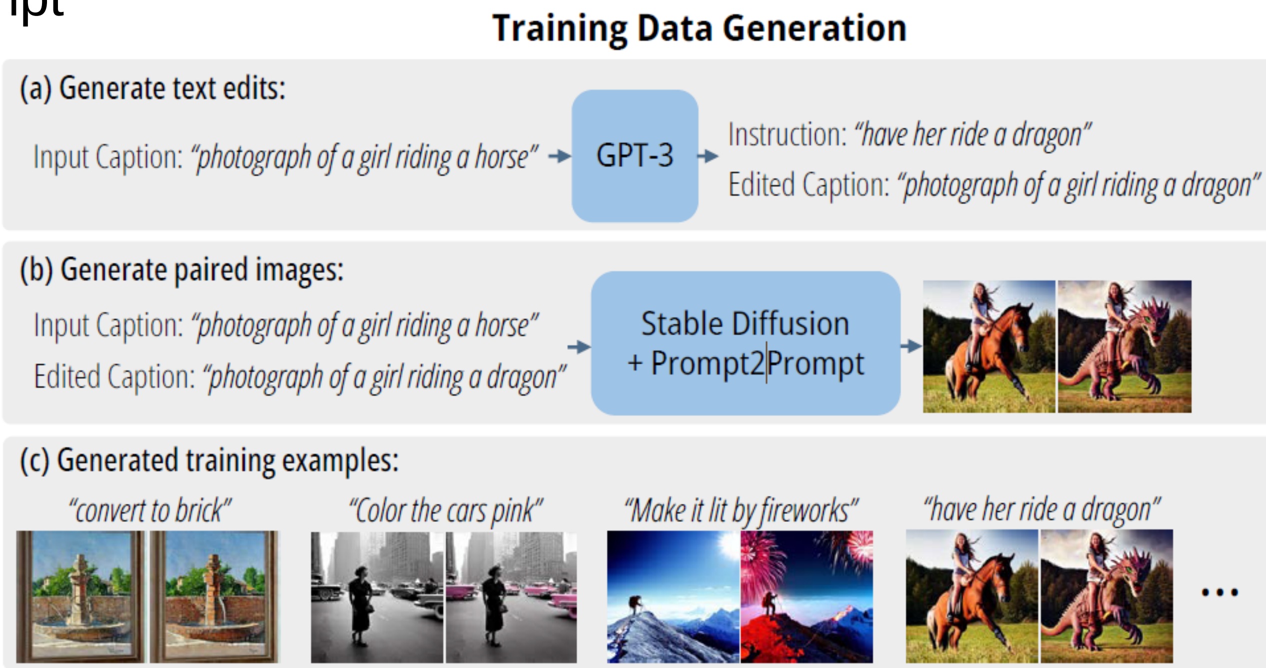
- Instruct Pix2Pix [Brooks-2023]
  - Textual editing or real images

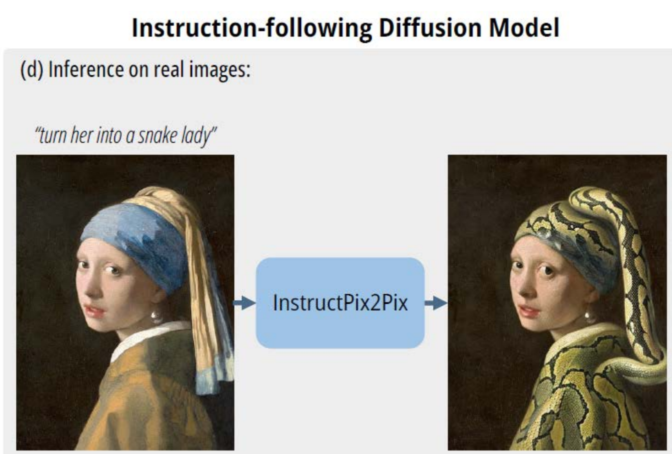


Instruct Pix2Pix

- New model trained from synthetic captions (Chat GPT) and images edited by Prompt-to-Prompt

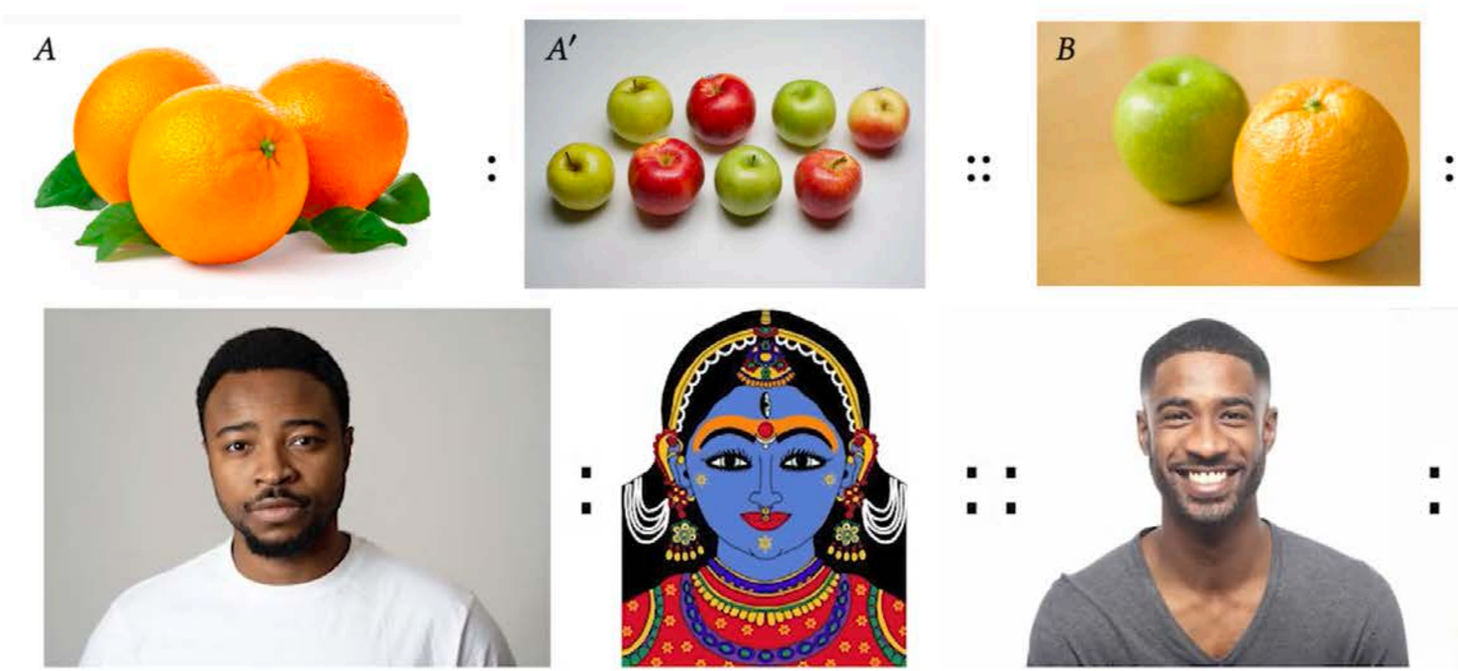
450k training examples





Diffusion Image Analogies

- Diffusion Image Analogies [Šubrtová-SIGGRAPH-2023]



?

inputs

output

- Real image editing by visual analogies
- Relation between B’ and B is analogous to relation between A’ and A
- Exploiting “algebra” of latent space of the Stable Diffusion

Diffusion Image Analogies

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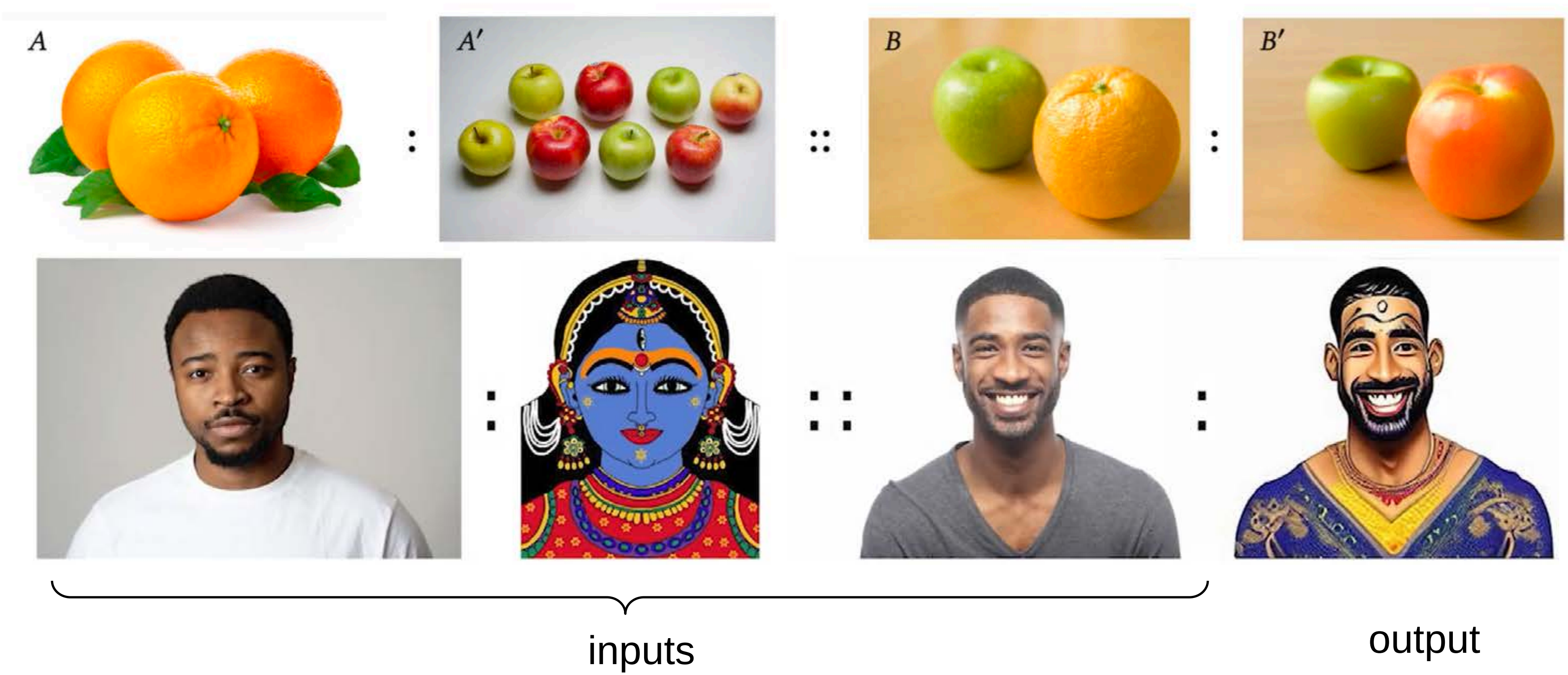

  - Real image editing by visual analogies
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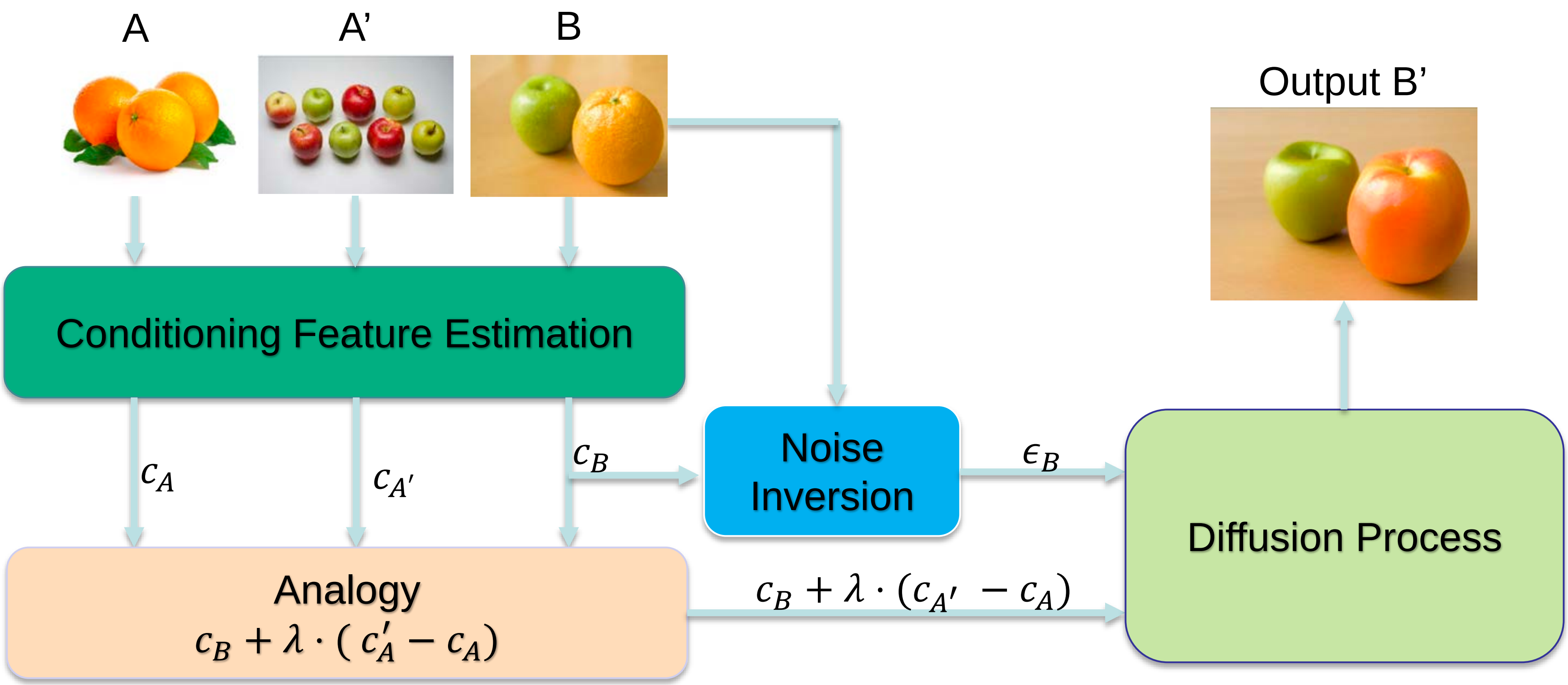
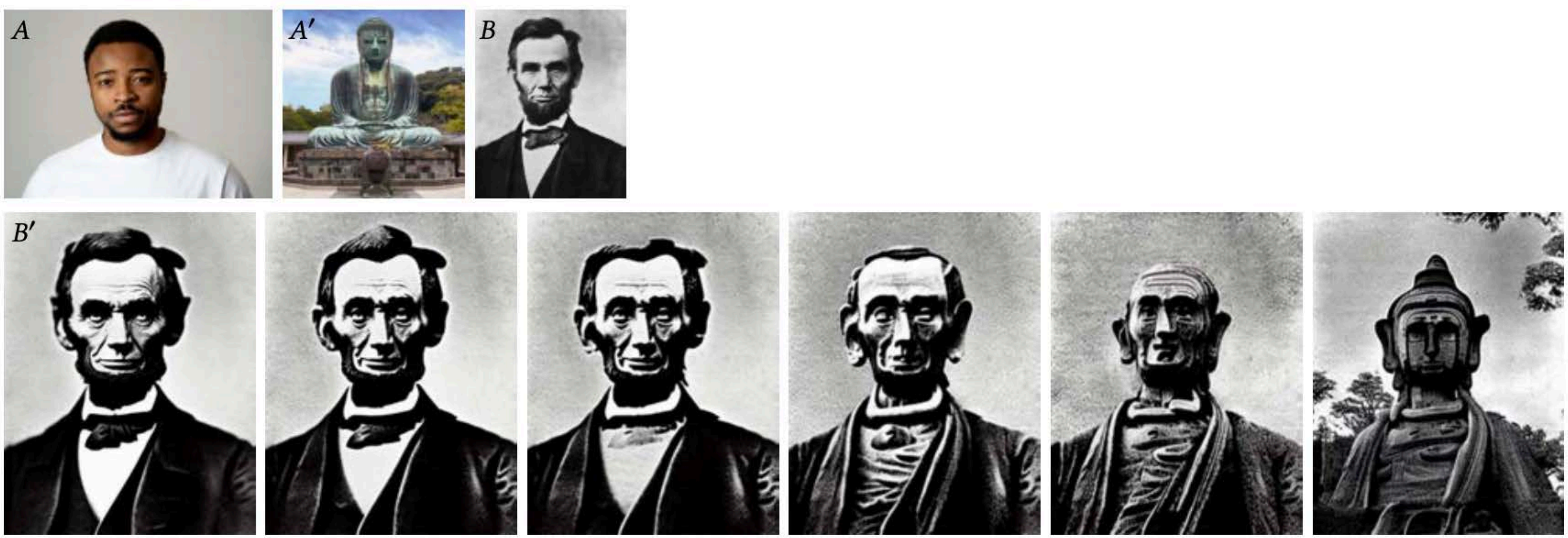
Diffusion Image Analogies



- Conditioning and Noise Inversion found by optimization
- Parameter λ controls the strength of the analogy

Diffusion Image Analogies

increasing analogy strength λ



increasing analogy strength λ

Diffusion models

PROS:

- **Excellent sample quality and mode coverage (state of the art)**
- Flexible conditioning (text, images, masks,…)
- Meaningful latent space (interpolation, semantic directions)
- Many models available
- Many image editing approaches

CONS:

- Slow sampling – iterative denoising
- Computationally expensive to train
- Memory heavy
- Approximate likelihood

Summary and Comparison

Summary and comparison

- Several approaches:
  1. Variational Autoencoders [Kingma-2014]
  2. Generative Adversarial Networks (GANs) [Goodfellow-2014]
  3. Autoregressive models [Oord-2016]
  4. Normalizing flow models [Dinh-2017]
  5. Diffusion models [Sohl-Dickstein-2015, Rombach-2022]

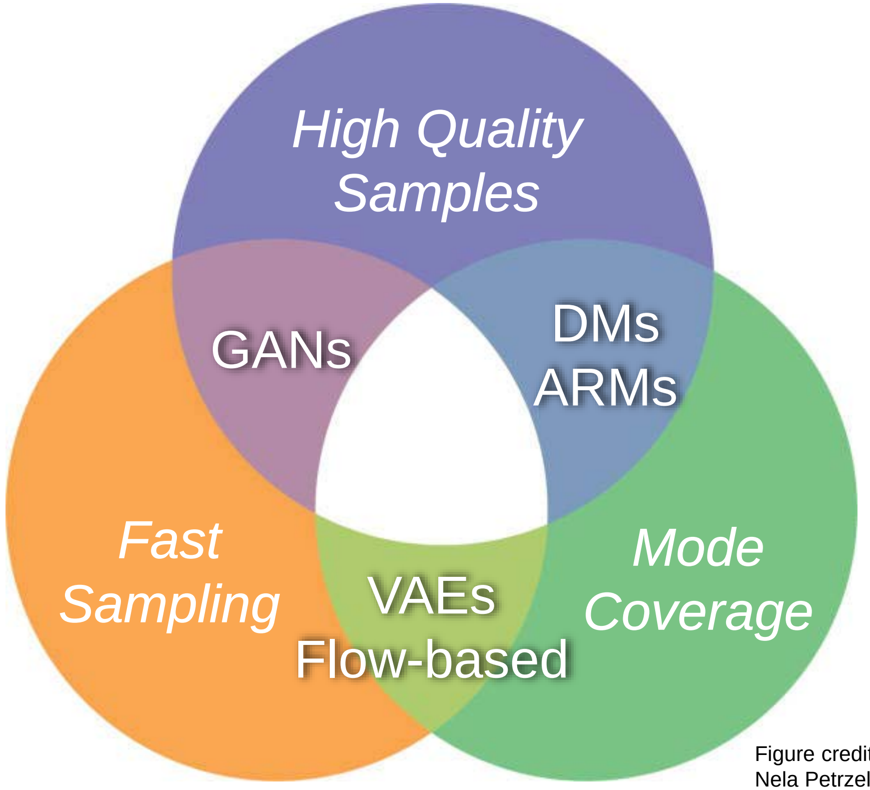


Figure credit
Nela Petrzelkova

Summary and Comparison

	VAE	GAN	Autoregressive	Normalizing Flow	Diffusion Model
Likelihood	Approximate	None	**Exact**	**Exact**	Approximate
Sampling Speed	**Fast**	**Fast**	Slow (sequential)	**Fast**	Slow (iterative denoising)
Training Stability	**Stable**	Unstable (adversarial)	**Stable**	**Stable**	Stable but expensive
Expressivity/quality	Good	Excellent	Excellent	Good	**State-of-the-art**
Latent Space	Yes (smooth)	Yes (implicit)	No	Yes (invertible)	Yes (via noise levels)
Density Estimation	Approximate	Impossible	Excellent	Excellent	Approximate
Inversion ($x \rightarrow$ latent)	Easy (encoder)	Hard / ill-posed	Not applicable	**Exact**	Approximate
Main Bottleneck	Blurry samples	Hard training, Mode collapse	Slow sampling	Low res / Latent space dimensionality	Long sampling chain

Conclusions

- "Photorealistic" quality has been achieved in the last couple of years
  - GANs, Diffusion models
- Deep generative models have evolved dramatically recently
- Interest / Fear of artistic community
  - Many tools greatly support content creativity
  - Certain creative artists feel endangered and exploited (some models likely trained on data without author's permission)
- Threat of high-quality deep fakes easy to perpetrate
  - Fake news, fake porn, etc.